

h2

Question 1

```
x <- 3
y <- 4
# a
a1 <- log(x + y) * x * y
a1
```

[1] 23.35092

```
# b
a2 <- log10((x*y)/2)
a2
```

[1] 0.7781513

```
# c
a3 <- (2 * x^(1/3)) + (y^(1/4))
a3
```

[1] 4.298713

```
# d
a4 <- (10^(x-y)) + exp(x*y)
a4
```

[1] 162754.9

Question 2

```
x <- c(4, 6, 5, 7, 10, 9, 4, 15)
# a
b1 <- x < 7
b1
```

[1] TRUE TRUE TRUE FALSE FALSE FALSE TRUE FALSE

```
# b
b2 <- x == 7
b2
```

```
[1] FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE
```

```
# c
x = 7

b3 <- x
b3
```

```
[1] 7
```

```
# <=
# >=
# !=

# d
x <- c(4, 6, 5, 7, 10, 9, 4, 15)

mean(x)
```

```
[1] 7.5
```

```
a <- sum(x)
a
```

```
[1] 60
```

```
b <- length(x)
b
```

```
[1] 8
```

```
a/b
```

```
[1] 7.5
```

```
avg <- sum(x)/length(x)
avg
```

```
[1] 7.5
```

```
# d

sub_vector <- x[x > avg]

# e
```

```
max_subvector <- max(sub_vector)
max_subvector
```

```
[1] 15
```

Question 3

```
# a
A <- matrix(c(1, 3, 2, 5, 4, 3, 4, 4), nrow = 2, ncol = 4)
A
```

```
      [,1] [,2] [,3] [,4]
[1,]     1     2     4     4
[2,]     3     5     3     4
```

```
colnames(A) <- c("Col1", "Col2", "Col3", "Col4")
A
```

```
      Col1 Col2 Col3 Col4
[1,]     1     2     4     4
[2,]     3     5     3     4
```

```
# b
B = A + 3
B
```

```
      Col1 Col2 Col3 Col4
[1,]     4     5     7     7
[2,]     6     8     6     7
```

```
# c
A + B
```

```
      Col1 Col2 Col3 Col4
[1,]     5     7    11    11
[2,]     9    13     9    11
```

```
# d
D <- cbind(A, c(1,2))
D
```

```
      Col1 Col2 Col3 Col4
[1,]     1     2     4     4 1
[2,]     3     5     3     4 2
```

```
# e
D[ , 3]
```

```
[1] 4 3
```

Question 4

```
# a
LETTERS[18]
```

```
[1] "R"
```

```
#b
LETTERS[length(LETTERS)]
```

```
[1] "Z"
```

```
#c
Names <- c("John", "Andrew", "Thomas")
Designation <- c("Manager", "Project Head", "Marketing Head")

cbind(Names, Designation)
```

```
      Names    Designation
[1,] "John"    "Manager"
[2,] "Andrew"  "Project Head"
[3,] "Thomas"  "Marketing Head"
```

```
cbind(c("John", "Andrew", "Thomas"),
      c("Manager", "Project Head", "Marketing Head"))
```

```
      [,1]    [,2]
[1,] "John"   "Manager"
[2,] "Andrew" "Project Head"
[3,] "Thomas" "Marketing Head"
```

```
# d
friend <- c("Peter", "John")
city <- c("Abuja", "Warri")

paste(friend,"lives in", city)
```

```
[1] "Peter lives in Abuja" "John lives in Warri"
```

```
#e
```

Question 5

```
#a
```

```
names <- c("Bob", "Alice", "Jane")
age <- c(22, 65, 36)
flag <- c(FALSE, FALSE, TRUE)

df <- data.frame(names, age, flag)
df
```

```
  names age  flag
1   Bob  22 FALSE
2  Alice  65 FALSE
3   Jane  36  TRUE
```

```
# b
```

```
df$score <- c(50, 60, 90)

df
```

```
  names age  flag score
1   Bob  22 FALSE    50
2  Alice  65 FALSE    60
3   Jane  36  TRUE    90
```

```
# c
```

```
## add a row
new_name = "Tom"
new_age = 43
new_flag = FALSE
new_score = 67

new_row <- data.frame(names = new_name,
                      age = new_age,
                      flag = new_flag,
                      score = new_score)

new_row <- data.frame(names = "Tom",
                      age = 43,
                      flag = FALSE,
                      score = 67)

new_row
```

```
  names age  flag score
1   Tom  43 FALSE    67
```

```
new_df <- rbind(df, new_row)
new_df
```

```
  names age  flag score
1   Bob  22 FALSE    50
2  Alice  65 FALSE    60
3   Jane  36  TRUE    90
4    Tom  43 FALSE    67
```

Question 6

```
# a
View(airquality)

airquality$Date_character <- paste(airquality$Day,
                                   airquality$Month,
                                   "1973",
                                   sep = "-")

airquality$Date <- as.Date(paste(airquality$Day,
                                 airquality$Month,
                                 "1973",
                                 sep = "-"),
                          "%d-%m-%Y")

View(airquality)

# b

names(airquality)[4] <- "Temperature"
names(airquality)[2] <- "Solar Radiation"

names(airquality)[names(airquality) == "Ozone"] <- "ozone"

Temperature <- (airquality$Temperature - 32)/1.8
```

Question 7

```
# a
hot <- max(airquality$Temperature)
cold <- min(airquality$Temperature)
hot
```

[1] 97

```
cold
```

```
[1] 56
```

```
which.max(airquality$Temperature)
```

```
[1] 120
```

```
which.min(airquality$Temperature)
```

```
[1] 5
```

```
airquality$Date[which.max(airquality$Temperature)]
```

```
[1] "1973-08-28"
```

```
airquality$Date[which.min(airquality$Temperature)]
```

```
[1] "1973-05-05"
```

```
airquality$ozone[which.max(airquality$Temperature)]
```

```
[1] 76
```

```
airquality$Date[which.max(airquality$Wind)]
```

```
[1] "1973-06-17"
```

```
# b
mean(airquality$Wind)
```

```
[1] 9.957516
```

```
airquality$Wind > mean(airquality$Wind)
```

```
[1] FALSE FALSE TRUE TRUE TRUE TRUE FALSE TRUE TRUE FALSE FALSE FALSE
[13] FALSE TRUE TRUE TRUE TRUE TRUE TRUE FALSE FALSE TRUE FALSE TRUE
[25] TRUE TRUE FALSE TRUE TRUE FALSE FALSE FALSE FALSE TRUE FALSE FALSE
[37] TRUE FALSE FALSE TRUE TRUE TRUE FALSE FALSE TRUE TRUE TRUE TRUE
[49] FALSE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE
[61] FALSE FALSE FALSE FALSE TRUE FALSE TRUE FALSE FALSE FALSE FALSE FALSE
[73] TRUE TRUE TRUE TRUE FALSE TRUE FALSE FALSE TRUE FALSE FALSE TRUE
[85] FALSE FALSE FALSE TRUE FALSE FALSE FALSE FALSE FALSE TRUE FALSE FALSE
[97] FALSE FALSE FALSE TRUE FALSE FALSE TRUE TRUE TRUE FALSE TRUE TRUE
[109] FALSE FALSE TRUE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE
```

```
[121] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE TRUE TRUE TRUE TRUE
[133] FALSE TRUE TRUE FALSE TRUE TRUE FALSE TRUE TRUE TRUE TRUE FALSE TRUE
[145] FALSE TRUE TRUE TRUE FALSE TRUE TRUE FALSE TRUE
```

```
sum(airquality$Wind > mean(airquality$Wind))
```

```
[1] 72
```

```
#c
complete.cases(airquality)
```

```
[1] TRUE TRUE TRUE TRUE FALSE FALSE TRUE TRUE TRUE FALSE FALSE TRUE
[13] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[25] FALSE FALSE FALSE TRUE TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE
[37] FALSE TRUE FALSE TRUE TRUE FALSE FALSE TRUE FALSE FALSE TRUE TRUE
[49] TRUE TRUE TRUE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
[61] FALSE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE
[73] TRUE TRUE FALSE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE FALSE
[85] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE FALSE
[97] FALSE FALSE TRUE TRUE TRUE FALSE FALSE TRUE TRUE TRUE TRUE FALSE TRUE
[109] TRUE TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE TRUE FALSE TRUE
[121] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[133] TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE TRUE
[145] TRUE TRUE TRUE TRUE TRUE FALSE TRUE TRUE TRUE
```

```
sum(complete.cases(airquality))
```

```
[1] 111
```

```
#d
airquality$Date[airquality$`Solar Radiation` > 300]
```

```
[1] "1973-05-04" NA NA NA "1973-05-16"
[6] "1973-05-17" "1973-05-19" "1973-05-22" NA "1973-06-10"
[11] "1973-06-14" "1973-06-15" "1973-07-06" NA NA
[16] NA
```

```
airquality$Date[!is.na(airquality$`Solar Radiation`) &
  airquality$`Solar Radiation` > 300]
```

```
[1] "1973-05-04" "1973-05-16" "1973-05-17" "1973-05-19" "1973-05-22"
[6] "1973-06-10" "1973-06-14" "1973-06-15" "1973-07-06"
```

Question 9


```
climate <- read.csv("ncdc_galesburg_daily_climate.csv")  
  
View(climate)  
  
climate[climate == -9999] <- NA  
  
climate$DATE <- as.Date(as.character(climate$DATE),  
                        format =  
                        "%Y%m%d")
```